

Deepak Sathyan

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Education

Ph.D. University of Maryland

Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass

Advisor: Dr. Kaustubh Agashe

College Park, MD

Aug 2018 – July 2024

B.A., Suffolk University

Cum Laude Physics

Minor: Mathematics | GPA: 3.9

Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Boston, MA

Aug 2014 – May 2018

Employment

Mitchell Institute for Fundamental Physics and Astronomy Texas A&M University

Postdoctoral Research Associate

College Station, TX

Sept 2024 – Aug 2027

Awards

Breakthrough Prize in Fundamental Physics

Muon g-2 Collaboration

Apr 2026

Skills

High Energy Physics

Computing

Projects

BSM Searches via W Boson Mass

Developed a framework unifying beyond-the-Standard-Model searches with precision measurements of the W boson mass at the LHC.

Jan 2020 – July 2024

Muon g-2 Experiment

Contributed to the straw tracking detector and beam dynamics corrections for the Fermilab Muon g-2 experiment, which measured the muon anomalous magnetic moment to 0.46 ppm.

Aug 2018 – Jan 2022

Publications

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan

arxiv.org/abs/2605.08010

May 2026

A Baryon and Lepton Number Violation Model Testable at the LHC

A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan

arxiv.org/abs/2508.21064

Aug 2025

New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2507.01000

July 2025

New laboratory constraints on neutrinophilic mediators P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang arxiv.org/abs/2407.12738 (Phys. Lett. B)	Jan 2025
‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_{W} K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2404.17574 (JHEP)	Jan 2025
A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2310.13687 (Phys. Lett. B)	Jan 2024
Unifying Searches for New Physics with Precision Measurements of the W Boson Mass D. Sathyan	Jan 2024
Energy-peak based method to measure top quark mass via B-hadron decay lengths K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan arxiv.org/abs/2212.03929 (JHEP)	Jan 2023
TF07 Snowmass Report: Theory of Collider Phenomena Collaboration arxiv.org/abs/2210.02591	Oct 2022
Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021 Collaboration arxiv.org/abs/2209.13128	Sept 2022
Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021 Collaboration arxiv.org/abs/2209.11267	Sept 2022
Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan arxiv.org/abs/2204.02928	Apr 2022
Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan arxiv.org/abs/2203.13305	Mar 2022
The straw tracking detector for the Fermilab Muon g-2 Experiment Collaboration arxiv.org/abs/2111.02076 (JINST)	Jan 2022
Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm Muon g-2 arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)	Jan 2021

Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab Muon g-2 arxiv.org/abs/2104.03240 (Phys. Rev. Accel. Beams)	Jan 2021
LHC Signals for KK Graviton from an Extended Warped Extra Dimension K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan arxiv.org/abs/2008.06480 (JHEP)	Jan 2020

Selected Publications

Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm Muon g-2 arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)	Jan 2021
New laboratory constraints on neutrinophilic mediators P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang arxiv.org/abs/2407.12738 (Phys. Lett. B)	Jan 2025
LHC Signals for KK Graviton from an Extended Warped Extra Dimension K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan arxiv.org/abs/2008.06480 (JHEP)	Jan 2020